

ASSESSMENT TOOL - 1.

SHORT ANSWER TEST.

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Course: Computer Vision.

① Computer Vision:-

Computer vision is a part of Artificial Intelligence. It enables the computer to capture, process, analyze and image and videos. The computer will analyze the image by pixel values of intensity.

a) Main Objectives:-

- To acquire the images
- To analyze and understand object
- To detect & recognition
- To support the and make a decision.

b) Two Real World Application:-

- * Face detecting.
- * Self driving.

② Key Features of Computer Vision:-

The computer vision it extract the useful information from the images, analyze the image, recognition and understand the images and objects. It is extract the useful information from object by detecting images.

a) Feature extraction:

The feature extraction is an important of the computer vision it analyzed the image, edges, corners, textures shapes and colour of the images.

b) Recognition improve computer vision:

Recognition improves the features of the computer vision analysis by making the image to useful informations.

③. Different Levels of Image processing.

The Image processing have has Three They are

- Low-level [Removing noise]
- Mid-level [Detecting edges]
- High-level. [Understand the object Make Decisions].

a). Low-level processing:

Low-level image processing is gives and improves the image quality through operation

- Noise removal
- Filtering
- Contrast enhancement.

b). Different level image process mid level & High level.

Mid-level	High-level.
Mid-level the image process are gives Detecting edges and identity the object by analysis by pixel.	High-level image process means it gives Understand the image & make a proper decision

④ Digital Image and Fundamental Component?

The Digital Images represent the as the matrix format to analysis by the object and images. Each pixel stores the brightness or colour information.

a) pixel

A pixel (Picture element) is smallest unit cell intensity of the images and object by using to identify the image.

b) Image resolution on affect image Quality

Higher resolution on affect image Quality means more pixels, resulting the clearer and sharper images. Lower resolution produced blur images. It not in clear image.

⑤ How does image sensing & Acquisitions Real World to digital images?

Image sensing and Acquisitions convert real world and digital images using camera & sensors.

a) Role of Image sensor?

Role of Image sensor capture the light and convert into a digital image by using the CCD, CMOS, etc.

b) Two image acquisition devices.

Digital camera
Webcam.

⑥ Sampling & Quantization

Sampling converts the images into pixels, while Quantization assign intensity values to each of the pixels.

a) Define Sampling:

Sampling is define as the process of dividing a continuous images into discrete pixels.

b) Purpose Quantization:

Quantization converts continuous intensity values into discrete digital values of storage and processing.

⑦ Capturing Images?

Capturing images is it explain about the light from scene passes through a lens and form an image on the sensor.

a) perspective projection map a 3D scene onto a 2D images while perspective dept perception.

b) Illumination provides light that determines the brightness of the image and contrast the image in the good Quality.

Image processing & Computer Vision.

Image processing is an enhancement images, while computer vision understand image content.

a) Image processing is the technique of improving and modify the digital images.

Image processing	Computer vision.
Enhance Image	Understand the Image
Output is an improvement image	Output is information images.

9. Application of Computer Vision:

Computer vision widely used in many real-world applications

a) Application healthcare.

It helps detect the disease like medical images such as X-rays, CT scanners, MRI scanner.

b). Autonomous Vehicle:

It detects road, traffic sign, predicting and other vehicles for a safe driving.

⑩ Computer Vision Fundamentals.

Computer Vision of Fundamentals are analyzing the image quality and produced the good quality of images. Basic concepts of images.

- a). Because every digital images are made up of pixels. and Brightness of the images.
- b). Sampling determines the number of pixel. resolution determines the image clarity. High sampling and resolution improve the image quality.